C291 - Spring 25 - Systems Programming with C & Unix

Assignment 3

Q.1 Domain Names (20 marks)

Write a C program to generate and print all possible two-letter domain names. You can assume that these names are NOT case sensitive, so AA.com would be the same as aa.com. For this reason, only use lowercase letters. You should print each domain name on a new line, or in some other formatted fashion that results in a readable output.

Hint: In C, you can conveniently increment characters by their Ascii value. For example, if variable 'letter' is equal to 'a', then writing 'letter++' will increment the variable to value 'b'.

Output Example:

www.aa.com
www.ab.com
www.ac.com
.
. *(all other possible combinations of 2 letters)*
.
www.zx.com
www.zy.com
www.zz.com

Note: The solution for this problem should be named as solution_1.c and it should be under C291-Spring-25/assignment3 where assignment3 would be the new directory you would create.

Q.2 Prime Factorization (20 marks)

Write a C program to compute the Prime Factorization of a given number. You should ask the user for an integer, compute the results, and print the result.

Note: You are not expected to hold the entire result in some data type, then print. It is instead suggested that you print as you compute.

Note: Integers 0 and 1 have no prime factors. In the case the user enters one of these, print a special message stating this fact. Additionally, we can assume that only positive integers will be input.

Example 1:
Enter a positive integer that you wish to find prime factors of: 147
Output: The prime factorization of 147 is: 3 7 7.

Example 2:
Enter a positive integer that you wish to find prime factors of: 79
Output: The prime factorization of 79 is: 79. ← *This is because 79 is a prime number itself.*

Example 3:
Enter a positive integer that you wish to find prime factors of: 1
Output: 1 has no prime factorization.

Example 4:
Enter a positive integer that you wish to find prime factors of: 8
Output: The prime factorization of 8 is: 2 2 2.

Note: The solution for this problem should be named as solution_2.c and it should be under C291-Spring-25/assignment3 where assignment3 would be the new directory you would create.

Q.3 Reverse Fizz-Buzz (20 marks)

Fizz Buzz is a classic programming problem that often appears in interviews and similar environments. The basics of Fizz Buzz are as follows:

- Starting at 1, print integers in order.
- If the number is divisible by 3, instead of printing the number, print 'Fizz'.
- If the number is divisible by 5, instead of printing the number, print 'Buzz'.
- If the number is both divisible by 3 and 5, print 'Fizz-Buzz'.

For this question, you are to implement a slightly modified version of this. You are to ask the user for the starting number, and starting at that number, you will do Fizz Buzz in **reverse order**. That is, you will compute and print correct output starting at the given number, down to 1.
Additionally, when printing, **only 4 (or less, if all that is left) calculations should be on each line**.

Note: The program should only accept positive, non-zero integers. Other inputs should be given an error.

Example 1:
Enter the starting Fizz Buzz number: 15
Output:
Fizz-Buzz 14 13 Fizz
11 Buzz Fizz 8
7 Fizz Buzz 4
Fizz 2 1

Example 2:
Enter the starting Fizz Buzz number: -1
Output: Error: Cannot compute Fizz Buzz of -1.

Note: The solution for this problem should be named as solution_3.c and it should be under C291-Spring-25/assignment3 where assignment3 would be the new directory you would create.

Q.4 Reverse Binary (20 marks)

Write a C program to print the reverse of the binary representation of a given integer. Reverse binary means that the binary number is read in reverse (1010 -> 0101). You should take input from the user to determine the number used for conversion.

As the output is the reverse of the binary representation, your output should exclusively be composed of 0's and 1's. If you are unfamiliar with binary numbering, find more info here. This problem **does not** need bit manipulation in C.

We are not concerned with negative integers, so the input should only be in the form of positive integers.

Note: You are not expected to hold this result in some data type. It is acceptable to print the result as you compute. Also, we will ignore the leading zeroes, i.e. $(18)_{10}$ is $(00010010)_2$ in 8-bit representation but we will ignore the first 3 0s from the left, so $(18)_{10}$ is $(10010)_2$ and the reverse will be $(01001)_2$

Example 1:
Enter a number to find the reverse binary representation: 18
Output: 01001

Explanation: $(18)_{10} = (10010)_2$ and the reverse is $(01001)_2$

Example 2:
Enter a number to find the reverse binary representation: 224
Output: 00000111

Explanation: $(224)_{10} = (11100000)_2$ and the reverse is $(00000111)_2$

Example 3:
Enter a number to find the reverse binary representation: -40
Output: Error: Number must be a positive integer.

Note: The solution for this problem should be named as solution_4.c and it should be under C291-Spring-25/assignment3 where assignment3 would be the new directory you would create.

Q.5 Diamond Star (20 marks)

Write a program to print Diamond Star pattern in C. You should accept the number of rows from the user.

Note: You should only accept odd, positive numbers for this. Even numbers will not produce the diamond pattern, and negative numbers won't work for obvious reasons.

Expected output:
Enter the number of rows: 5

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    *
   ***
  *****
   ***
    *
```

Note: The solution for this problem should be named as solution_5.c and it should be under C291-Spring-25/assignment3 where assignment3 would be the new directory you would create.